

Rayan Baderdine

Computer & Communication Engineer

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SUMMARY

Accomplished professional with expertise in software development, embedded systems engineering, and data science. Proficient in Python, scikit-learn, XGBoost, and SHAP to build and deploy end-to-end machine learning pipelines. Committed to leveraging technical skills and engineering knowledge to drive innovation and deliver impactful technology solutions.

WORK EXPERIENCE

Data Science Intern | Codveda Technology — Remote

Mar 2026 — April 2026

- Completed 6 end-to-end data science projects across 3 difficulty levels within 1 month.
- Built a web scraper using BeautifulSoup & requests to collect and store structured data (CSV/JSON).
- Performed full data cleaning & preprocessing: missing value imputation, outlier removal (IQR), encoding, and normalization using pandas and scikit-learn.
- Developed regression models (Linear Regression, Decision Tree, Random Forest) — $R^2 = 0.89$ on Boston Housing dataset.
- Built a multi-class classifier (Logistic Regression, SVM, Random Forest) achieving 96.7% accuracy on Iris dataset.
- Implemented NLP sentiment classification using TF-IDF vectorization, Naive Bayes & Logistic Regression.
- Designed a feed-forward neural network (MLP, 3 hidden layers) with backpropagation & hyperparameter tuning — 96.7% test accuracy.
- Tools: Python, scikit-learn, pandas, matplotlib, NLTK, BeautifulSoup, NumPy.

Tutor | Self-employed

Oct 2024 — Jul 2025

- Clarified complex concepts during sessions to improve student comprehension and increase overall academic performance.
- Designed customized lesson plans addressing individual learning styles and promoted sustained engagement throughout the course.
- Monitored student progress regularly to identify knowledge gaps and adapt teaching methods for consistent improvement.

Member in Tutoring Academy (Summers) | Summer Academy

Jun 2018 — Present

- Contributed curriculum adaptation to diverse learning styles, increasing student confidence and fostering a more inclusive educational environment.
- Monitored student progress during sessions to identify learning gaps, enabling timely support that enhanced overall academic performance.
- Engaged students in interactive problem-solving activities that strengthened critical thinking skills and increased enthusiasm for challenging subject matter.

PROJECTS

Customer Churn Prediction with SHAP Explainability

2026

Python · XGBoost · SHAP · Streamlit · Optuna · scikit-learn

- Built an end-to-end ML pipeline on IBM Telco dataset (~7,000 customers, 20 features).
- Compared Logistic Regression, Random Forest, and XGBoost using 5-fold cross-validation.
- Tuned XGBoost with Optuna achieving 0.8485 AUC-ROC and 0.5873 F1 score.
- Applied SHAP to identify contract type and tenure as top churn drivers — enabling actionable business recommendations.
- Deployed as a live interactive web app using Streamlit. GitHub: github.com/ryan-baderdine/churn-prediction

EDUCATION

Master in Computer & Communication Engineering | Lebanese International University

Oct 2025 — Jun 2027

- Developed a Driver Fatigue Detection System (RISC-V, FreeRTOS)
- Developed a noise deletion audio filter using MATLAB

Bachelor of Science in Communication Engineering | Lebanese International University

Oct 2021 — Jun 2024

- Developed a website for mayors in Lebanon as a senior project
- Spectrum Analyzer
- Library Management System (SQL)
- BJT differential amplifier
- Guess the magical number (microcontroller)
- 4-bit binary adder
- Smart blind stick (Arduino)

SKILLS

Java, C, C++, C#, Python, Octave, HTML, CSS, JavaScript, PHP, SQL, Assembly, .NET, Azure, Android Studio, Arduino, AVR Microcontroller, Proteus, TinkerCAD, Atmel Studio, LTSpice, NetBeans, Visual Studio, Packet Tracer, MySQL, MATLAB, RISC-V, scikit-learn, pandas, matplotlib, NLTK, BeautifulSoup, NumPy, XGBoost, SHAP, Streamlit, Optuna.

COURSES

- HTML, CSS, and JavaScript for Web Developers — John Hopkins University (Jun 2025)
- Build a Free Website with WordPress — Coursera (Jun 2025)
- Fundamentals of Machine Learning and Artificial Intelligence — Amazon Web Services (May 2025)
- Foundational C# with Microsoft — Microsoft (Aug 2025)
- Azure Data Scientist Training — Microsoft (Mar 2026)

LANGUAGES

Arabic (Native) • English (C2 — Proficiency)

REFERENCES

Available upon request.

LINKS

TinkerCAD: <https://www.tinkercad.com/users/3g2vVRJzZDn?type=circuits>

LinkedIn: <https://www.linkedin.com/in/ryan-baderdine-5b32992a2>

GitHub: <https://github.com/ryan-baderdine>